Cluster Image Portal — Architecture & Sequences

Web GUI component of `imgctl` v2.2.0 · NVIDIA BCM / DGX head node

The one idea: a **root producer** collects images on a timer and atomically writes a JSON **snapshot**; an **unprivileged web service** only *reads* that snapshot. They share one file and nothing else — so the public, no-auth web tier never runs `imgctl`, never SSHes, and never sees Harbor credentials.

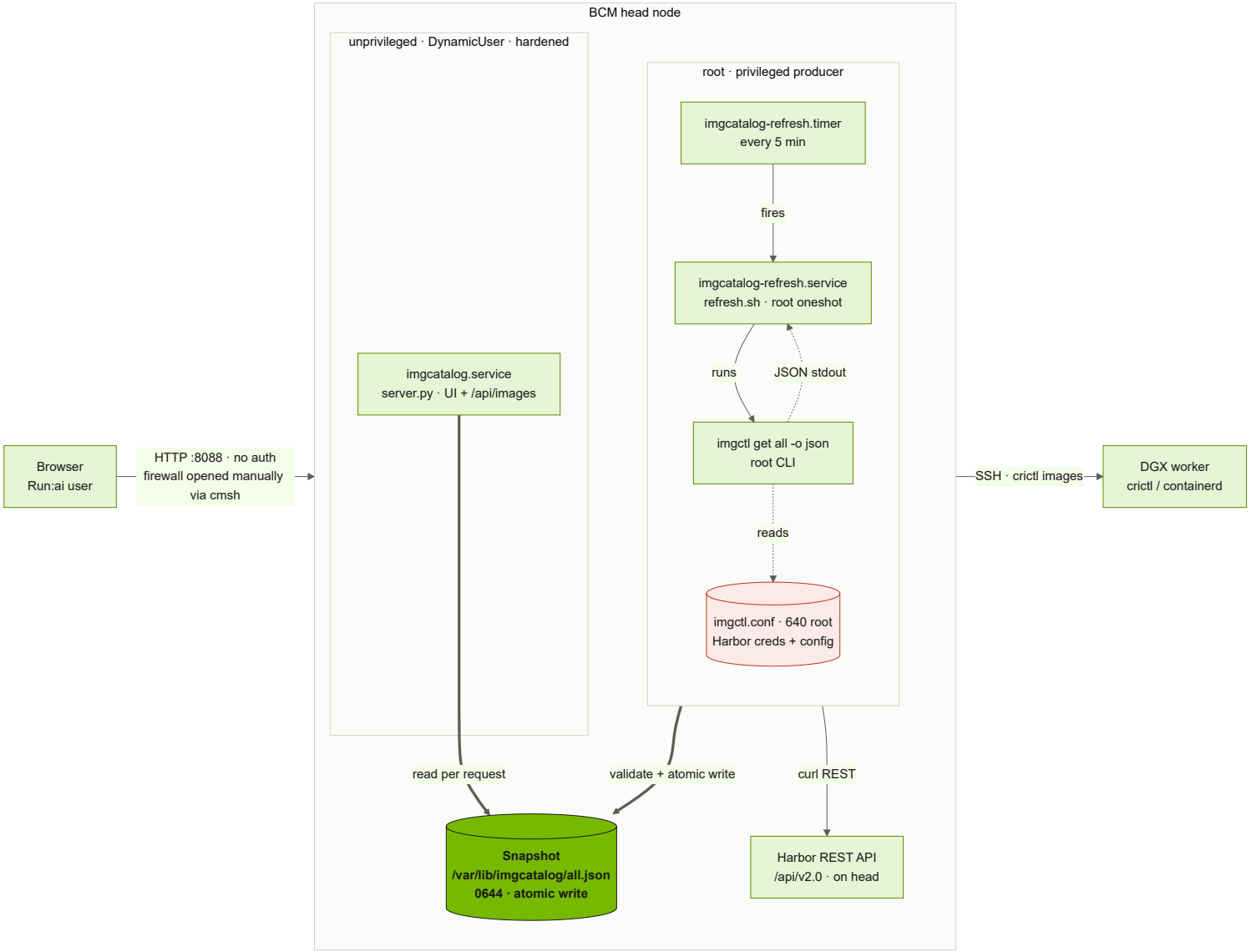
Components & responsibilities

Component	Runs as	Triggered by	Reads	Writes	Purpose
`imgcatalog-refresh.timer`	systemd	boot + every 5 min	—	—	Fires the refresh oneshot (`Persistent=true`)
`refresh.service` → `refresh.sh`	**root** (oneshot)	the timer	`imgctl.conf`	the snapshot	Run imgctl, validate, atomically publish
`imgctl`	**root** (CLI)	`refresh.sh`	`imgctl.conf`	stdout → temp	Query Harbor API + `crictl` over SSH
Harbor REST API	service on head	imgctl	—	—	Source: custom/registry images
DGX worker `crictl`	on worker (via SSH)	imgctl	—	—	Source: node-cached NGC/Docker images
Snapshot `all.json`	file (`0644`)	—	by web tier	by `refresh.sh`	The decoupling seam (one JSON file)
`imgcatalog.service` → `server.py`	**unprivileged** `DynamicUser`	always-on (`Restart=always`)	the snapshot	—	Serve UI + JSON API
Browser	client	user	`/` , `/api/images`	—	Browse + copy pull reference

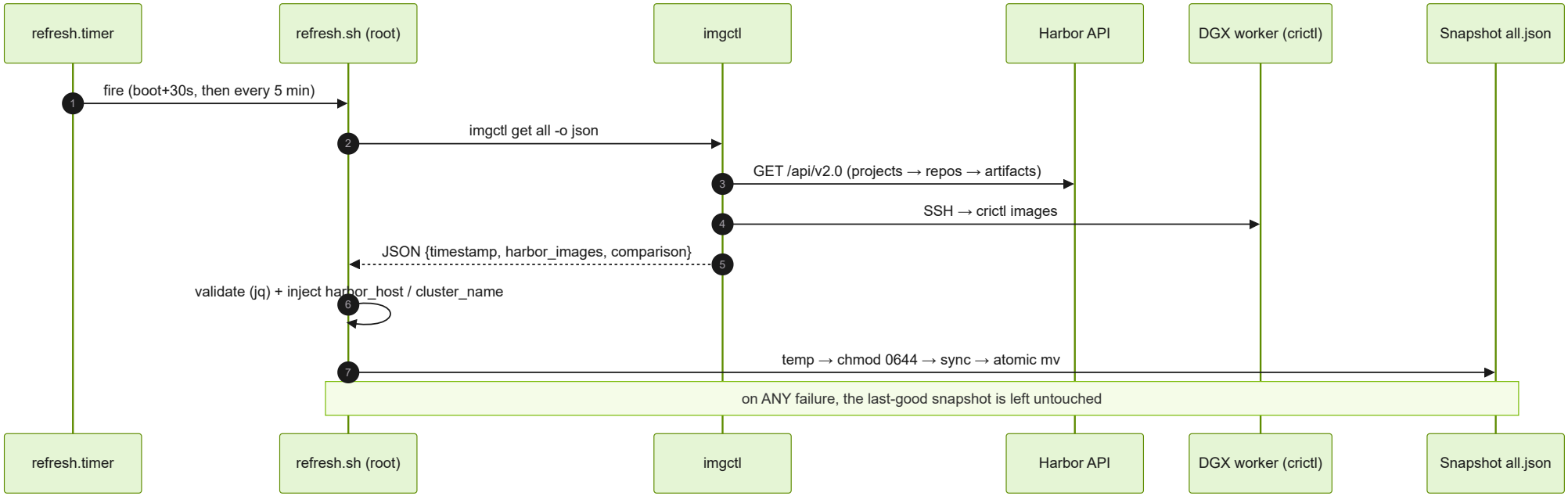
API: `GET /` (UI) · `GET /api/images` (flattened JSON, exact `reference` per image) · `GET /api/raw` · `GET /healthz` · `GET /version` . Non-GET/HEAD → `405` . | **Ops:** snapshot `/var/lib/imgcatalog/all.json` ; install via `install.sh` ; firewall opened **manually** via `cmsh … openports` (never hand-edit `/etc/shorewall/rules`).

Security: no authentication; binds `0.0.0.0:8088` (plain HTTP) — access gated **only by the firewall port**. The web tier is sandboxed (`DynamicUser` , `NoNewPrivileges` , empty `CapabilityBoundingSet` , `ProtectSystem=strict` , `MemoryMax` / `CPUQuota` / `TasksMax`) and read-only — it exposes image names/tags/sizes only.

1 · Architecture



2 · Sequence — refresh path (root producer, periodic)



3 · Sequence — request path (unprivileged web tier)